AI-Based Career Profile Analysis, Skill Gap Prediction and Job Matching

I found a **2026 IEEE Access journal paper**, which fits your requirement much better than the conference paper we discussed earlier.

1. Base Paper Details

Paper Title:

Multiview Job–Resume Matching With Contrastive Learning and LLM-Based Paraphrase Augmentation

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The paper was published in IEEE Access on **2 July 2026** according to the University of Zagreb research repository.

Base Paper – IEEE Access

2. What does the Base Paper do?

The paper addresses the problem of **matching resumes with suitable job descriptions**. It uses NLP and deep-learning techniques to represent resumes and job descriptions and learn their semantic similarity.

The researchers use a **dual-encoder architecture** with BERT-based representations and contrastive learning to improve job–resume matching. The work also explores paraphrase augmentation using LLMs to improve the robustness of resume representations.

3. Methods/Techniques

- Natural Language Processing

- BERT
- Transformer-based representations
- Dual Encoder
- Contrastive Learning
- Text Embeddings
- Cosine Similarity
- LLM-based paraphrase augmentation
- Information Retrieval

4. Research Gap

The base paper primarily focuses on **matching a resume with suitable job opportunities**.

Our proposed project can extend this by adding:

Resume → Skill Extraction → Skill Gap Prediction → Job Role Prediction → Skill Recommendations → What-if Career Simulation

So instead of only answering:

“Which job matches this resume?”

our project also attempts to answer:

“Which skills is the candidate missing, which roles are suitable, and what skills could improve their profile?”

5. How our project is different

Base Paper	**Our Proposed Work**
Job–resume matching	Career profile analysis
Focuses on matching	Matching + skill-gap analysis
Uses semantic representations	NLP + ML prediction
Resume ↔ Job similarity	Candidate skills ↔ Job requirements
Job matching	Job-role prediction